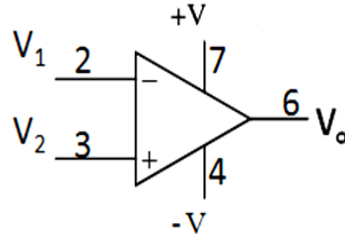
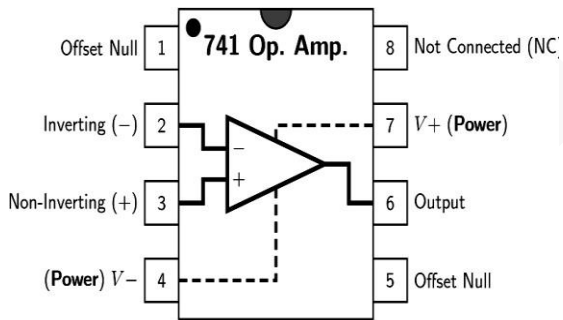


Module 2 : Part A - Operational amplifiers

The integrated-circuit *operational-amplifier* is the fundamental building block for many electronic circuits. An op-amp is a multi-stage, direct coupled, high gain negative feedback amplifier used to amplify AC and DC input signals.

- The main applications of op-amp: Active filters, oscillators, peak detector, comparators, voltage regulators, precision rectifiers, instrumentation and control systems, pulse generators, square wave generators etc.

PIN DIAGRAM OF μA 741



Pin 2: Inverting (- ve) terminal

Pin 3: Non-Inverting (+ve) terminal

Pin 4: - ve power supply ($-V_{EE}$)

Pin 5: +ve power supply ($+V_{CC}$)

Pin 6: Output terminal (V_{out})

Pins 1 & 5: Offset null and pin 8: not connected (NC)

Fig. 3.1 (a) Basic pin-out of Op-amp

(b) circuit symbol

(c) pin description

The '+' sign indicates zero phase shift while the '-' sign indicates 180° phase shift. Since 180° phase shift produces an inverted waveform, the '-' input is often referred to as the **inverting input**. Similarly, the '+' input is known as the **non-inverting input**.

Operational amplifier parameters

Open-loop voltage gain, (A_{ol}):

It is the internal voltage gain of the device and represents the ratio of output voltage to input voltage when there are no external components.

Ideal value: ∞ Practical Value: 2×10^5

The open-loop voltage gain is often expressed in decibels (dB) rather than as a ratio.

$$\text{Open loop gain} = 20 \log \frac{V_{out}}{V_{in}}$$

Most operational amplifiers have open-loop voltage gains of 90 dB or more.

Closed-loop voltage gain (A_{cl}):

The closed-loop voltage gain of an operational amplifier is defined as the ratio of output voltage to input voltage measured with a small proportion of the output fed-back to the input (i.e. with feedback applied). The effect of providing negative feedback is to reduce the loop voltage gain. Value of A_{cl} is very much less than value of A_{ol} .

$$\text{Closed loop gain} = \frac{V_{out}}{V_{in}}$$

Input resistance

The input resistance of an operational amplifier is defined as the ratio of input voltage to input current expressed in ohms.

Ideal value: ∞ Practical value: $2 \text{ M}\Omega$ for bipolar operational amplifiers and $10^{12} \Omega$ for CMOS operational amplifiers.

$$R_{IN} = \frac{V_{in}}{I_{in}}$$

Output resistance

The output resistance of an operational amplifier is defined as the ratio of open-circuit output voltage to short-circuit output current expressed in ohms.

Ideal value: 0, Practical values range from less than 10 Ω to around 100 Ω .

$$R_{OUT} = \frac{V_{out(OC)}}{I_{out(SC)}}$$

Input offset voltage

Practically, a small DC voltage will appear at the output of amplifier when no input voltage (or 0V) is applied. Thus, differential (very small) voltage is required between the inputs to make the output to 0V.

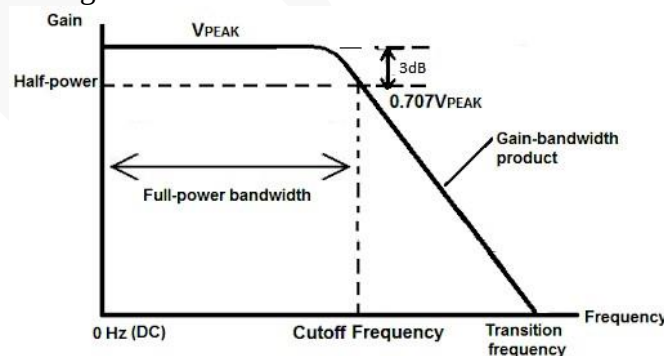
Input offset voltage may be minimized by applying relatively large amounts of negative feedback or by using the offset null facility provided by a number of operational amplifier devices.

Ideal value: 0 Typical values range from 1 mV to 15 mV.

Full-power bandwidth

It is the **range of frequencies** at which the maximum undistorted V_{peak} swing falls to 0.707 of its low-frequency (DC) value.

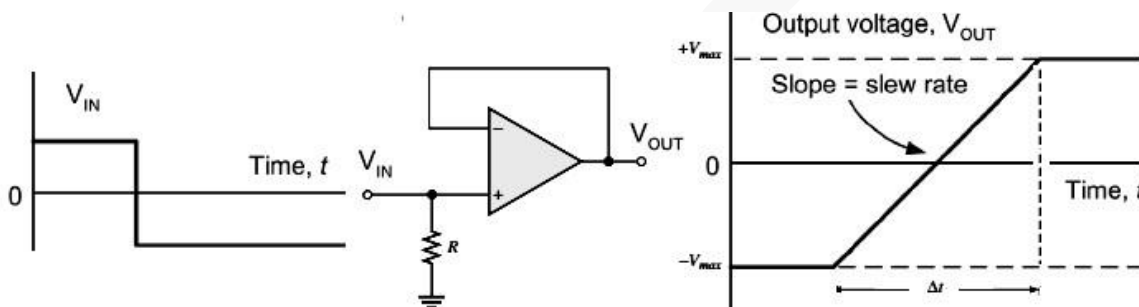
Typical full-power bandwidths range from 10 kHz to over 1 MHz



Slew Rate (SR)

The slew rate of an operational amplifier is the rate of change of output voltage with time in response to a perfect step-function input. Slew rate describes how fast the output voltage responds to an immediate change in input voltage. Slew rate is measured in V/s (or V/ μ s) and typical values range from 0.2 V/ μ s to over 20 V/ μ s.

$$SR = \left. \frac{dv_o}{dt} \right|_{\max} \quad V / \mu \text{ sec}$$



Operational amplifier characteristics

Characteristics for an 'ideal' operational amplifier are:

- (a) The open-loop voltage gain should be very high (ideally infinite).
- (b) The input resistance should be very high (ideally infinite).
- (c) The output resistance should be very low (ideally zero).
- (d) Full-power bandwidth should be as wide as possible.
- (e) Slew rate should be as large as possible.
- (f) Input offset should be as small as possible.

Comparison of operational amplifier parameters for 'ideal' and 'real' devices

Parameter	Ideal	Real
Voltage gain	Infinite	100,000
Input resistance	Infinite	100 MΩ
Output resistance	Zero	20 Ω
Bandwidth	Infinite	2 MHz
Slew rate	Infinite	10 V/μs
Input offset	Zero	Less than 5 mV

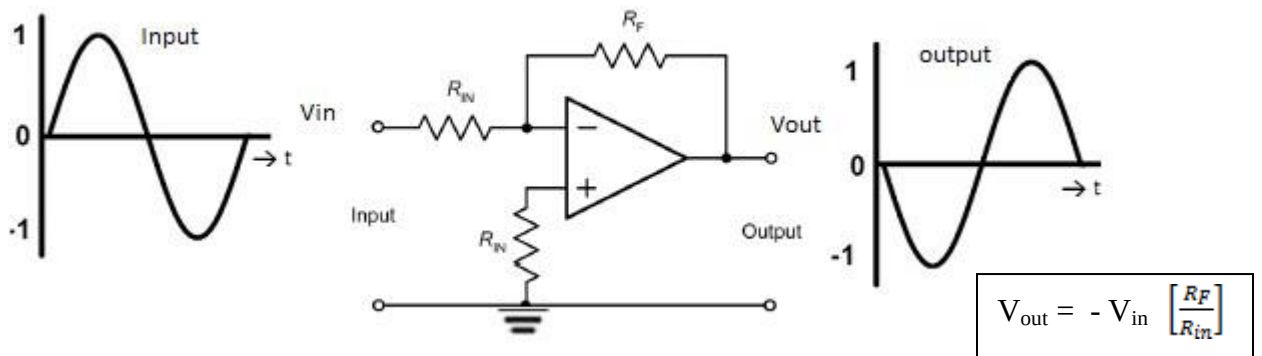
Some common examples of integrated circuit operational amplifiers

Device	Type	Open-loop voltage gain (dB)	Input bias current	Slew rate (V/μs)	Application
AD548	Bipolar	100 min.	0.01 nA	1.8	Instrumentation amplifier
AD711	FET	100	25 pA	20	Wideband amplifier
CA3140	CMOS	100	5 pA	9	Low-noise wideband amplifier
LF347	FET	110	50 pA	13	Wideband amplifier
LM301	Bipolar	88	70 nA	0.4	General-purpose operational amplifier
LM348	Bipolar	96	30 nA	0.6	General-purpose operational amplifier
TL071	FET	106	30 pA	13	Wideband amplifier
741	Bipolar	106	80 pA	0.5	General-purpose operational amplifier

Operational amplifier configurations

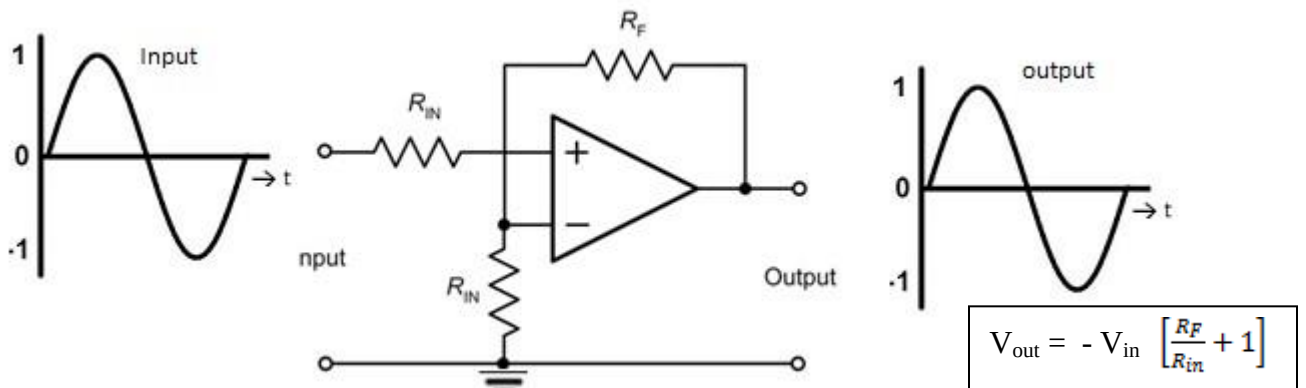
1) Inverting operational Amplifier

Input signal V_{in} is applied to the inverting terminal of the amplifier and output V_{out} is inverted version (180° phase shift) of input V_{in} .



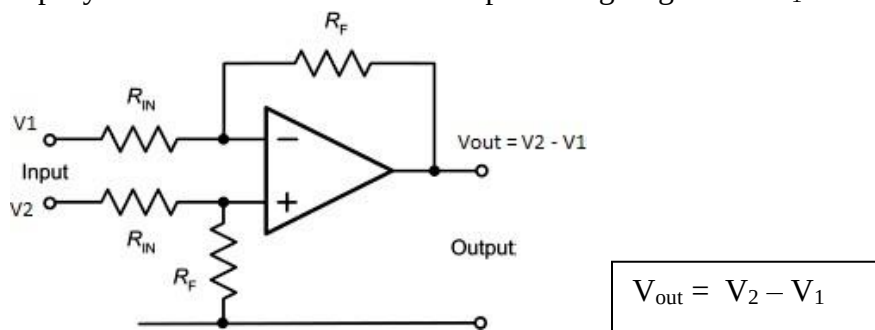
2) Non-inverting operational Amplifier

Input signal V_{in} is applied to the non-inverting terminal of the amplifier and output V_{out} is non-inverted version (0° phase shift) of input V_{in} .



3) Differential amplifiers

Differential amplifiers amplify the difference between two input voltage signals of V_1 and V_2 .



Effect of input and feedback capacitors

By selecting appropriate values of capacitor, the frequency response of an inverting operational voltage amplifier may be very easily modified to suit a particular set of requirements. The lower cut-off frequency is determined by the value of the input capacitance, C_{IN} , and input resistance, R_{IN} .

$$f_1 = \frac{1}{2\pi C_{IN} R_{IN}} = \frac{0.159}{C_{IN} R_{IN}}$$

The upper cut-off frequency will be determined by the feedback capacitance, C_F , and feedback resistance, R_F ,

$$f_2 = \frac{1}{2\pi C_F R_F} = \frac{0.159}{C_F R_F}$$

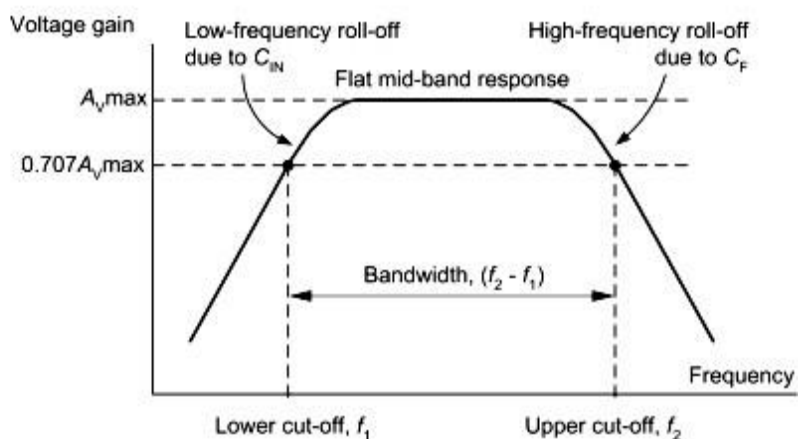
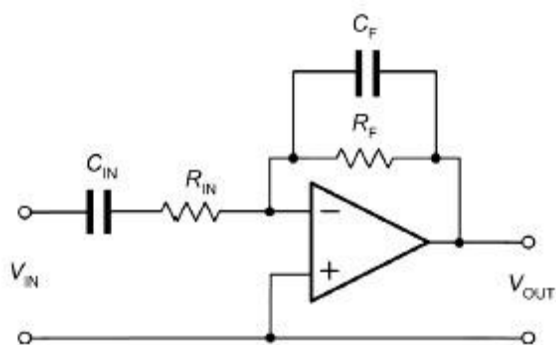
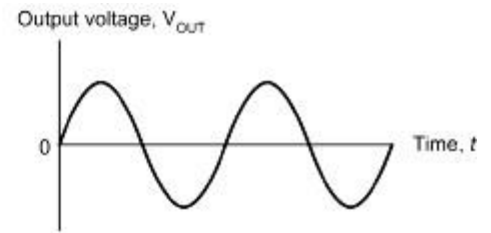
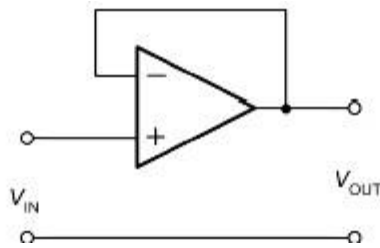
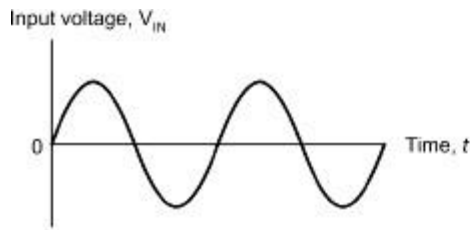


Fig.19 Effect of adding capacitors, C_{IN} and C_F , to modify the frequency response of an operational amplifier

Operational amplifier circuits

Voltage follower OPAMP

Output voltage V_{out} follows the input voltage V_{in} so the circuit is named as op-amp voltage follower. The output is connected directly back to the (-) inverting input so that the feedback is 100% and V_{in} is exactly equal to V_{out} . It is shown in the fig. If voltage V_{in} increases, voltage V_{out} increases. On the other hand, if voltage V_{in} decreases, voltage V_{out} also decreases. It provides an effective isolation of the output from the signal source that eliminating the loading effect of the second circuit from the first circuit.



Properties of Voltage follower

- Voltage gain = 1
- Input impedance $R_{in} = \infty$
- Output impedance $R_{out} = 0$
- Effective isolation of the output from the signal source.

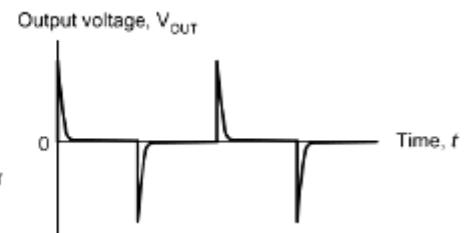
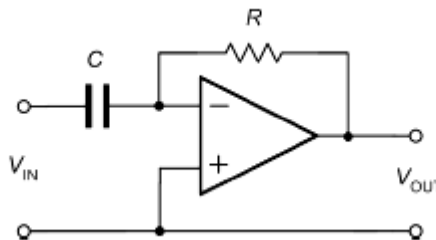
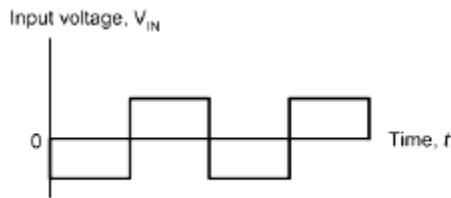
$$V_{out} = V_{in}$$

Differentiator amplifier

Differentiator produces output voltage (V_{out}) is proportional to the rate of change of the input voltage V_{in} . An op-amp differentiator is an inverting amplifier, which uses a capacitor C in series with the input voltage V_{in} and a feedback resistor R is connected between V_{out} and inverting (-) input.

$$V_{out} = -RC \frac{d}{dt}(V_{in})$$

If input is a square wave to a differentiator, output is pulses.

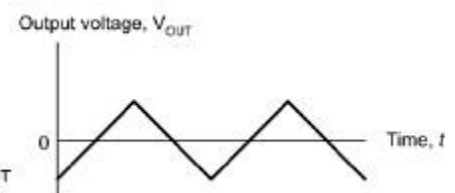
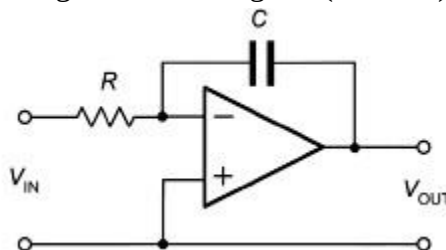
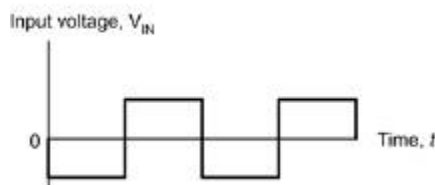


Integrator Amplifier

Integrator produces output voltage V_{out} , is proportional to the integral of the input voltage V_{in} . An op-amp *integrator* is an inverting amplifier, which uses a resistor R in series with the input voltage V_{in} and a capacitor C is connected between V_{out} and inverting (-) input as feedback.

$$V_{out} = -\frac{1}{CR} \int V_{in} dt$$

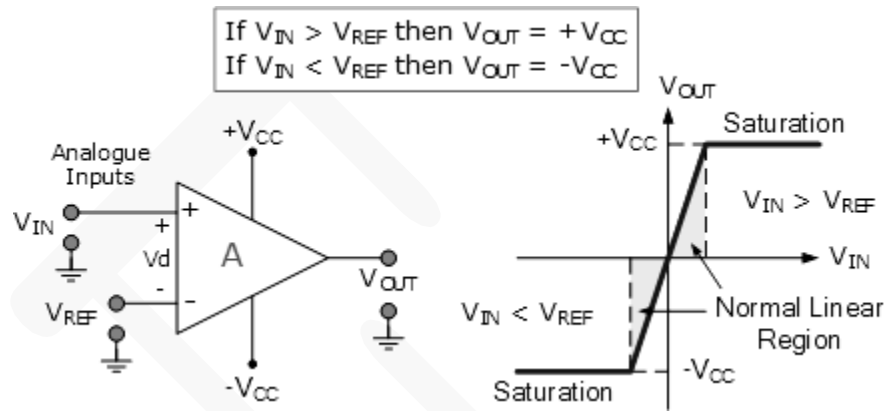
If input is a square wave, output of an integrator is a triangular (inverted) wave.



Comparator

OPAMP voltage comparator compares the magnitudes of two voltage inputs and determines which is the larger of the two.

- Referring the fig.3.11, assume ($V_{IN} < V_{REF}$).
- As the non-inverting (positive) input of the comparator is less than the inverting (negative) input, the output will be the negative supply voltage, $-V_{CC}$ resulting in a negative saturation of the output.
- When ($V_{IN} > V_{REF}$), the output voltage rapidly switches HIGH towards the positive supply voltage, $+V_{CC}$ resulting in a positive saturation of the output.



Voltage comparator using OPAMP

- Suppose the input voltage V_{IN} , is decreased slightly less than V_{REF} , the op-amp's output switches back to its negative saturation voltage acting as a threshold detector.
- Then it is seen that the op-amp voltage comparator is a device whose output is dependent on the value of the input voltages.

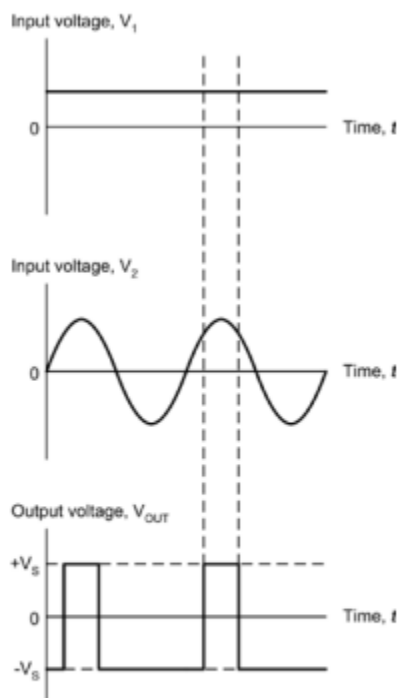


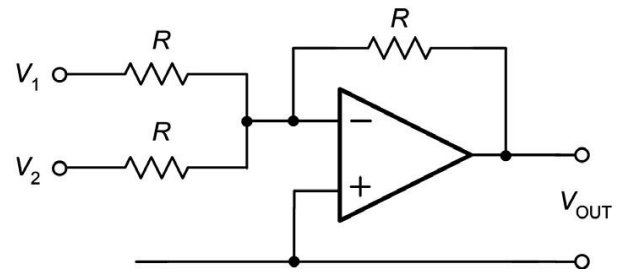
Figure 8.18 Typical input and output waveforms for a comparator

Summing Opamp

The inverting summing or adder op-amp circuit for three inputs is shown in the fig. 3.7.

The output voltage, V_{out} is proportional to the algebraic sum of the input voltages, V_1 , V_2 . Input signals V_1 and V_2 are applied to the inverting input of the op-amp through input resistors.

$$V_{out} = -(V_1 + V_2)$$



Inverting summing op-amp circuit

